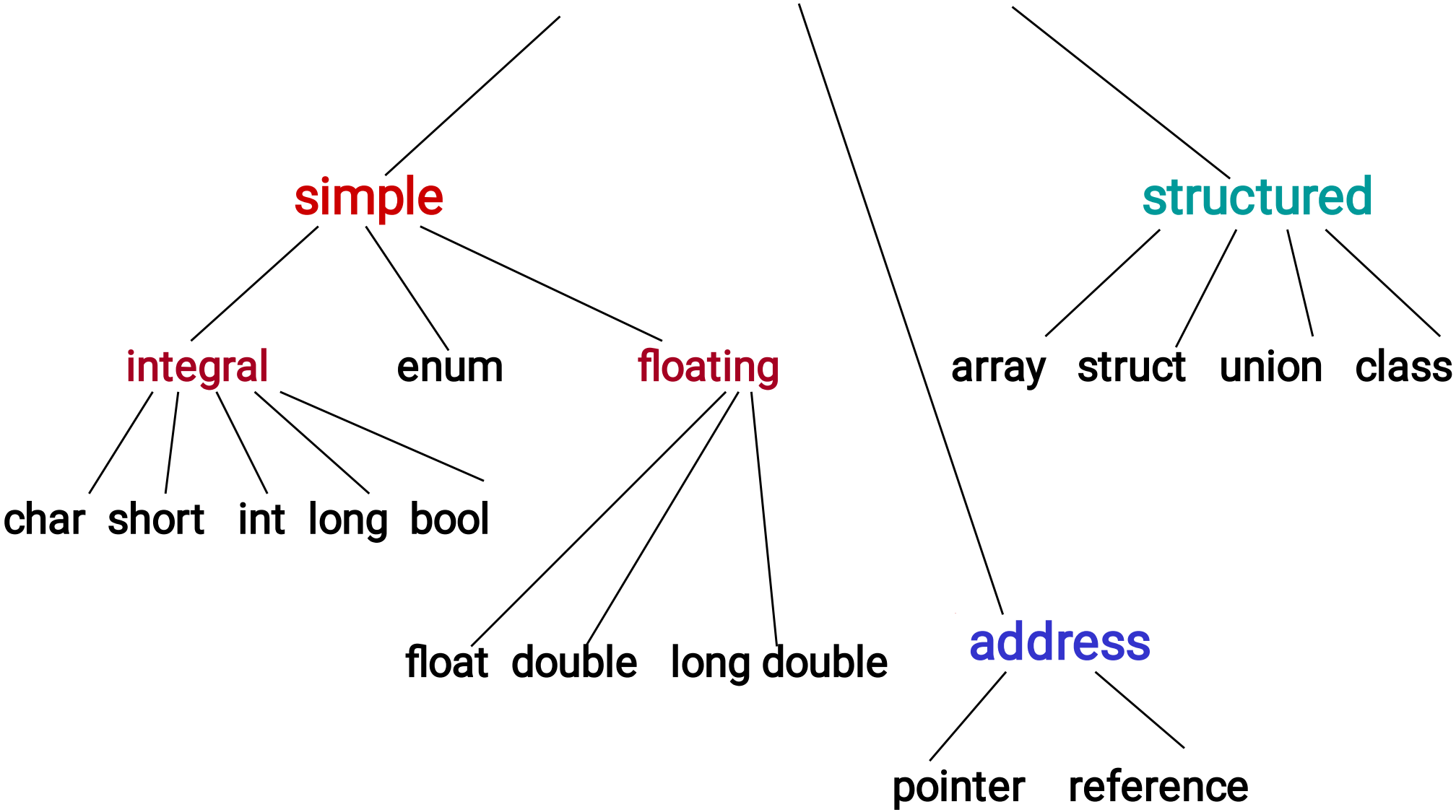
Pointers, Dynamic Data, and Reference Types

- Review on Pointers
- Reference Variables
- Dynamic Memory Allocation
  - The new operator
  - The delete operator
  - Dynamic Memory Allocation for Arrays

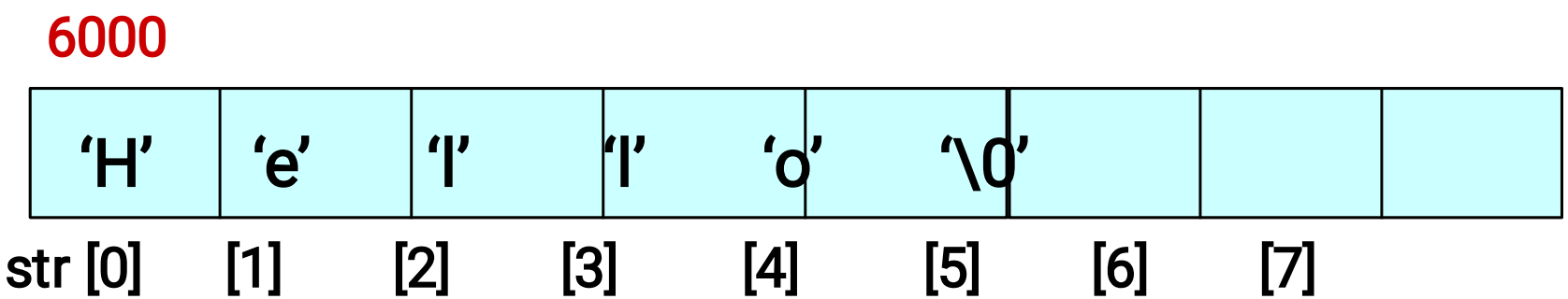
C++ Data Types


simple
structured
integral
enum
floating
array
struct
union
class
char
short
int
long
bool
float
double
long double
address
pointer
reference

Recall that . . .

```
char  str [ 8 ];
```

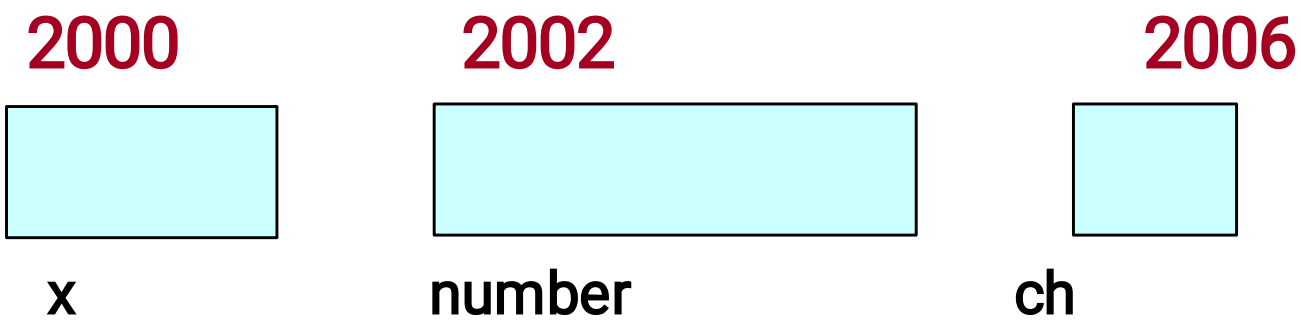
- **str** is the base address of the array.
- We say **str** is a pointer because its value is an address.
- It is a <u>pointer constant</u> because the value of **str** itself cannot be changed by assignment. It "points" to the memory location of a char.



Addresses in Memory

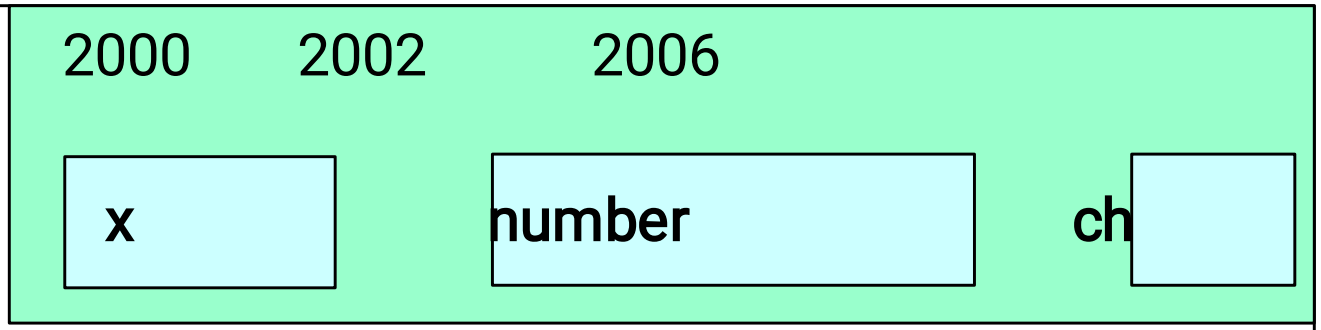
- When a variable is declared, enough memory to hold a value of that type is allocated for it at an unused memory location. This is the address of the variable

```
int     x;
float   number;
char    ch;
```



Obtaining Memory Addresses

- The address of a *non-array variable* can be obtained by using the address-of operator & (dereference operator)




```
int     x;
float   number;
char    ch;

cout << "Address of x is " << &x << endl;

cout << "Address of number is " << &number << endl;

cout << "Address of ch is " << &ch << endl;
```

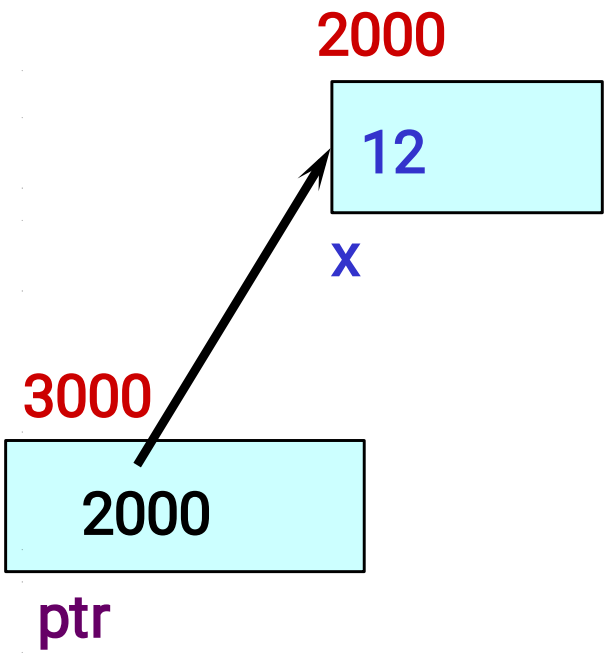
What is a pointer variable?

- A pointer variable is a variable whose value is the address of a location in memory.
- To declare a pointer variable, you must specify the type of value that the pointer will point to, for example,

```
int *ptr; // ptr will hold the address of an int

char *q;  // q will hold the address of a char
```

Using a Pointer Variable

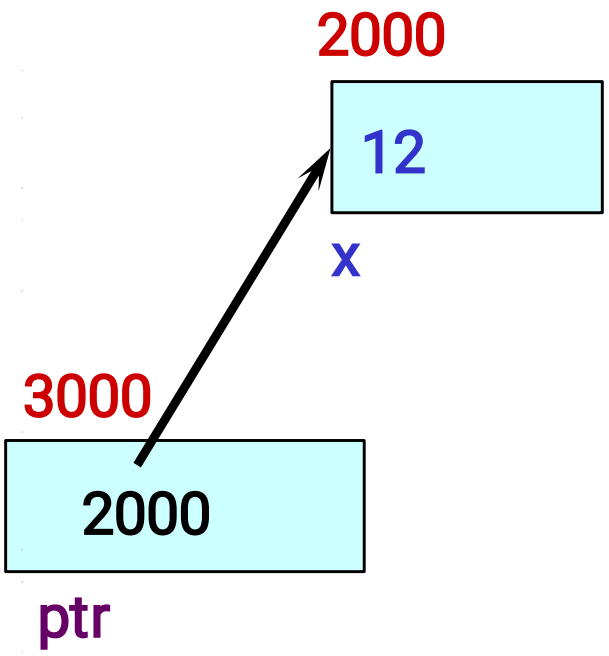
```
int  x;
x = 12;

int *ptr;
ptr = &x;
```



NOTE: Because ptr holds the address of x, we say that ptr "points to" x

*: reference operator

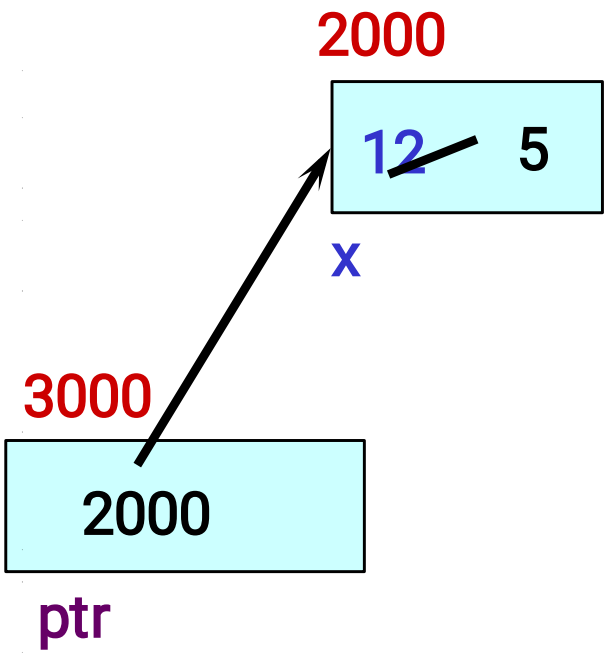
```
int  x;
x = 12;


int *ptr;
ptr = &x;

cout  <<*ptr;
```



NOTE: The value pointed to by ptr is denoted by *ptr

Using the Dereference Operator

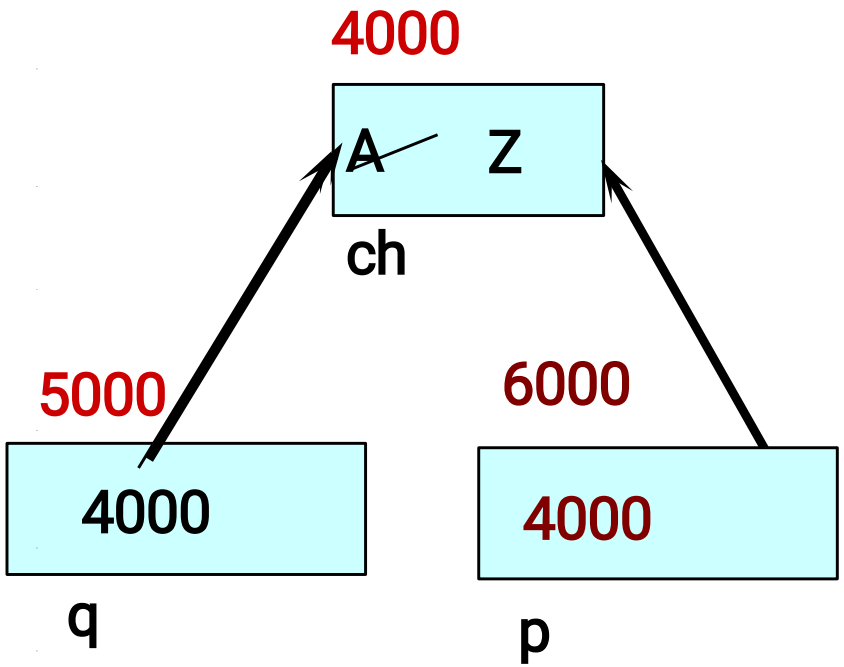
```
int  x;
x = 12;

int *ptr;
ptr = &x;

*ptr = 5;
```



// changes the value at the address ptr points to 5

Self –Test on Pointers

```
char  ch;
ch =  'A';

char *q;
q  = &ch;

*q = 'Z';
char *p;
p = q;
```

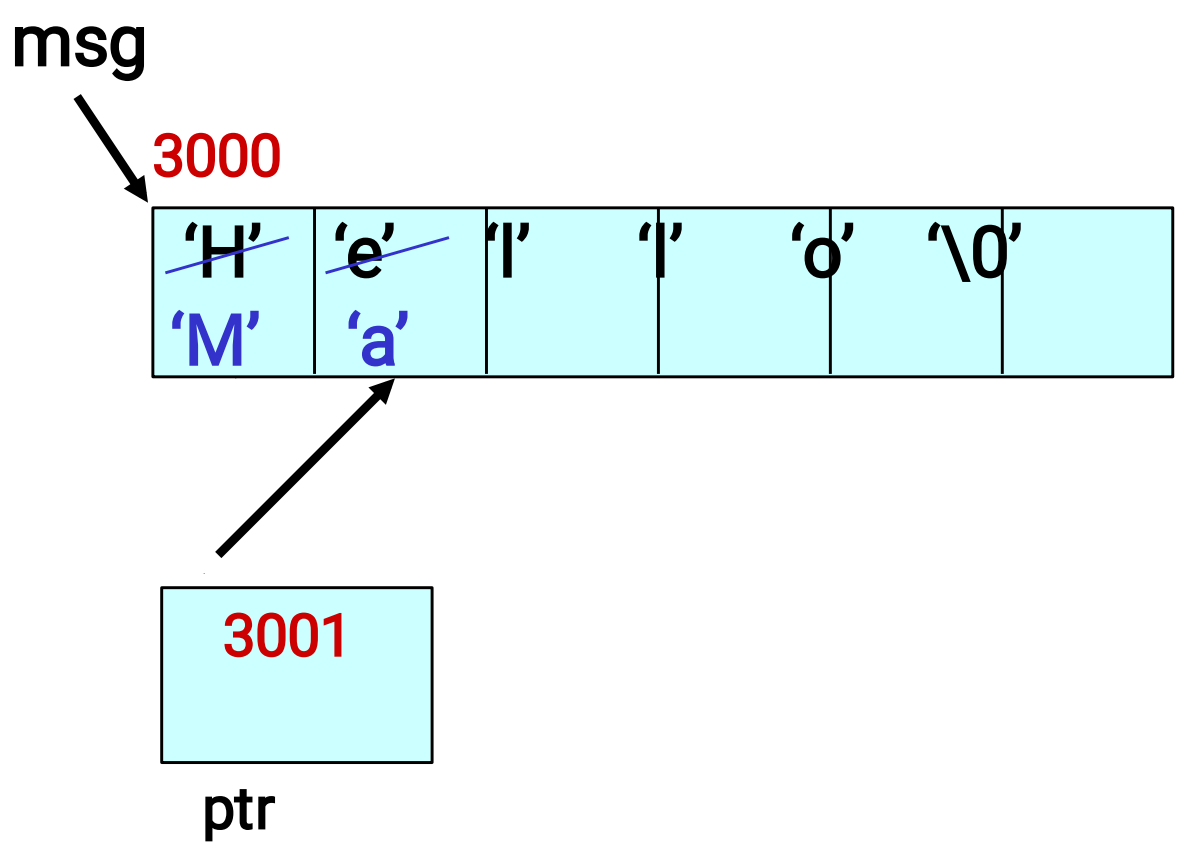


// the P has value 4000

// now p and q both point to ch

Using a Pointer to Access the Elements of a String

```
char   msg[ ] =“Hello”;
char *ptr;
ptr  =  msg;
*ptr  = ‘M’ ;
ptr++;
*ptr = ‘a’;
```



Reference Variables

Reference variable = ***alias for another variable***

- Contains the address of a variable (like a pointer)
- No need to perform any dereferencing (unlike a pointer)
- Must be initialized when it is declared

```
int x = 5;
int &z = x;     // z is another name for x
cout << x << endl; // prints 5
cout << z << endl;  // prints 5

z = 9;          // same as x = 9;

cout << x << endl;    // prints 9
cout << z << endl;    // prints 9
```

Why Reference Variables

- Are primarily used as function parameters

- Advantages of using references:
  - you don’t have to pass the address of a variable
  - you don’t have to dereference the variable inside the called function

Reference Variables Example

```
#include <iostream.h>

// Function prototypes (required in
    C++)

void p_swap(int *, int *);
void r_swap(int &, int &);

int main (void){
 int v = 5, x = 10;
 cout << v << x << endl;
 p_swap(&v,&x);
 cout << v << x << endl;
 r_swap(v,x);
 cout << v << x << endl;
 return 0;
}
```

```
void p_swap(int *a, int *b)
{
  int temp;
  temp = *a;
  *a = *b;
  *b = temp;
}
```

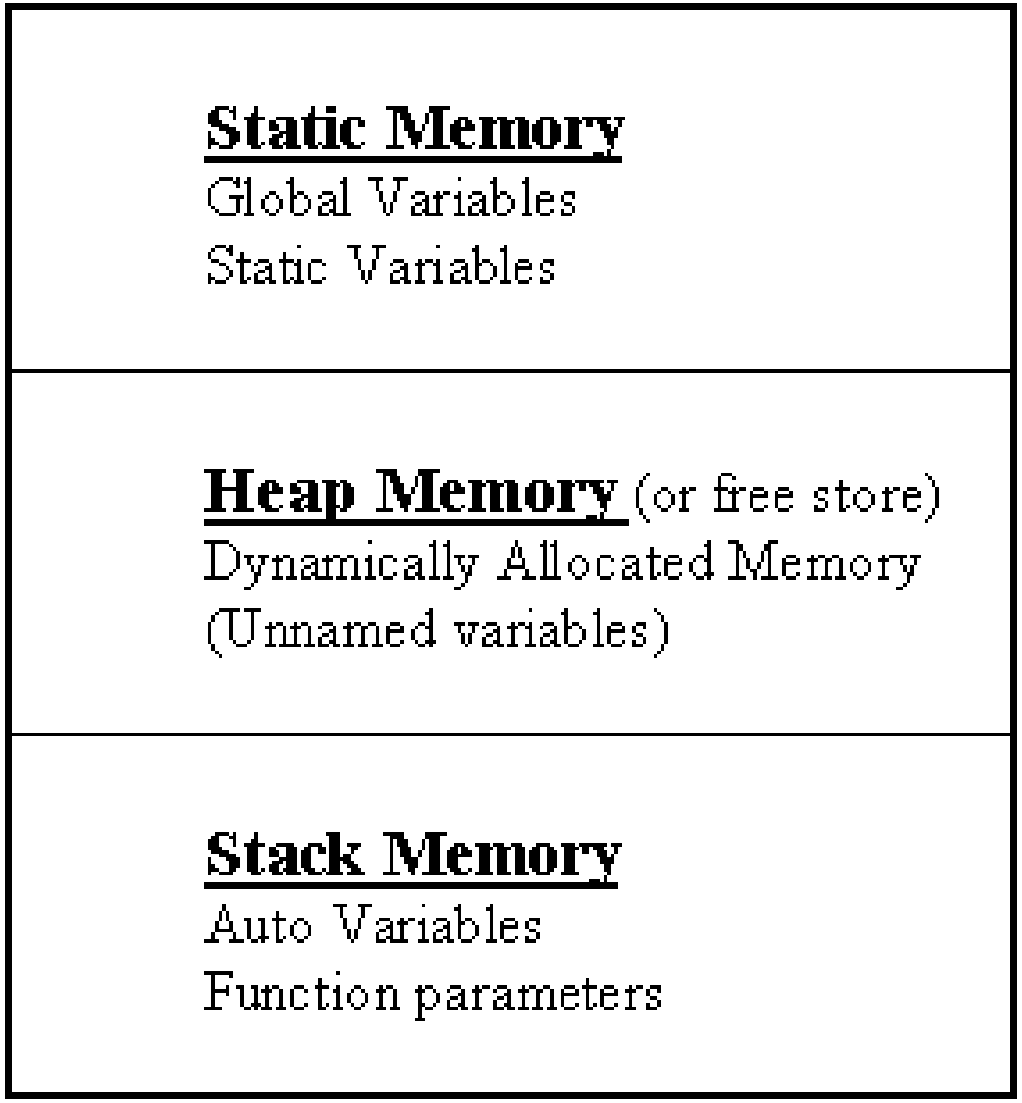
```
void r_swap(int &a, int &b)
{
 int temp;
 temp = a;
 a = b;
 b = temp;
}
```

Dynamic Memory Allocation

In C and C++, three types of memory are used by programs:

- **Static memory** - where global and static variables live
- **Heap memory** -dynamically allocated at execution time

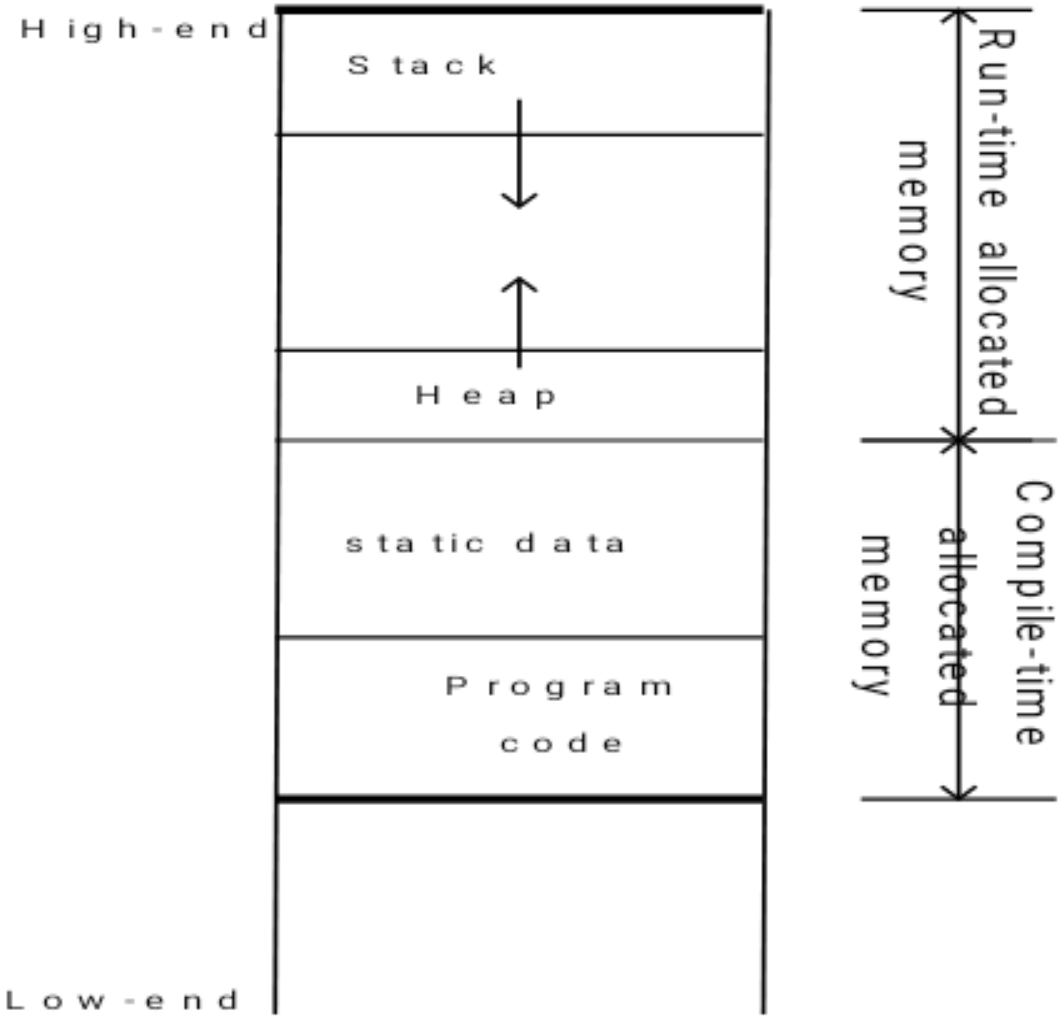
  -"small" amount of "managed" memory accessed using pointers
- **Stack memory** - used by automatic variables



3 Kinds of Program Data

- STATIC DATA: Allocated at compiler time
- DYNAMIC DATA: explicitly allocated and deallocated during program execution by C++ instructions written by programmer using operators **new** and **delete**
- AUTOMATIC DATA: automatically created at function entry, resides in activation frame of the function, and is destroyed when returning from function

Dynamic Memory Allocation Diagram



Dynamic Memory Allocation

- *In C++,* this is accomplished using the **new** and **delete** operators
- **new** is used to allocate memory during execution time
  - returns a pointer to the address where the object is to be stored
  - always returns a pointer to the type that follows the **new**

Operator **new** Syntax

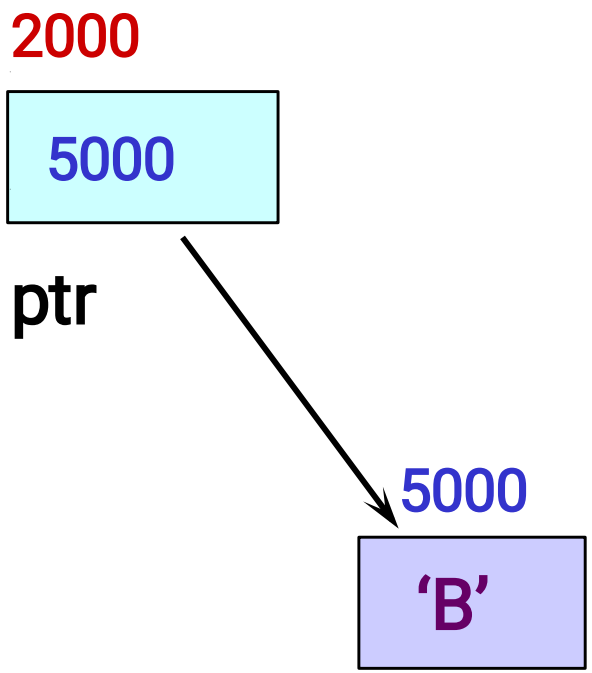
```
new   DataType
```

```
new   DataType  [IntExpression]
```

- If memory is available, in an area called the heap (or free store) new allocates the requested object or array, and returns a pointer to (address of) the memory allocated.
- Otherwise, program terminates with error message.
- The dynamically allocated object exists until the delete operator destroys it.

Operator new

```
char *ptr;

ptr = new char;
*ptr = 'B';
cout  <<  *ptr;
```



NOTE: Dynamic data has no variable name

The NULL Pointer

- There is a pointer constant called the “null pointer” denoted by NULL
- But NULL is not memory address 0.

- NOTE: It is an error to dereference a pointer whose value is NULL. Such an error may cause your program to crash, or behave erratically. It is the programmer’s job to check for this.

```
while (ptr != NULL) {
 . . .          // ok to use *ptr here
}
```

Operator **delete** Syntax

```
delete   Pointer
```

```
delete  []   Pointer
```

- The object or array currently pointed to by Pointer is deallocated, and the value of Pointer is undefined. The memory is returned to the free store.
- Good idea to set the pointer to the released memory to NULL
- Square brackets are used with delete to deallocate a dynamically allocated array.

Operator **delete**

```
char *ptr;


ptr = new char;

*ptr = 'B';

cout  <<  *ptr;
```

```
delete  ptr;
```

2000

???

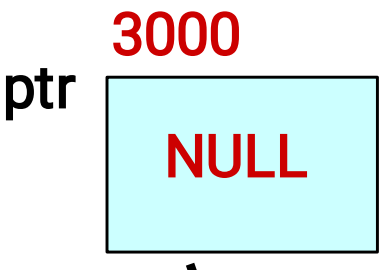
ptr

NOTE:

delete deallocates the memory pointed to by ptr

Example

```
char  *ptr ;

ptr  =  new  char[ 5 ];

strcpy( ptr, “Bye” );

ptr[ 1 ] = ‘u’;

delete  ptr;

ptr = NULL;
```



// deallocates the array pointed to by ptr
// ptr itself is not deallocated
// the value of ptr becomes undefined

Take Home Message

- Be aware of where a pointer points to, and what is the size of that space.
- Have the same information in mind when you use reference variables.
- Always check if a pointer points to NULL before accessing it.